

*** Macroscopic Balance for Momentum

*Assumption of perfect mixing obviates the need for a momentum balance in well-mixed reactors.

*For tubular reactors, mixing is not perfect in the axial direction. The momentum balance in an infinitesimal shell of thickness dz is

$$\begin{aligned} \frac{d}{dt} \int_{\Omega_z} \rho V_z dV - \int_{\partial\Omega_z} v_z (\rho V_z) dA + \int_{\partial\Omega_z+dz} v_z (\rho V_z) dA \\ = \int_{\partial\Omega_z} (\rho) dA + \int_{\partial\Omega_z+dz} (\rho) dA + \int_{\partial\Omega} dA \cdot P_z \end{aligned}$$

At steady state,

$$-\langle \rho v_z^2 \rangle_z + \langle \rho v_z^2 \rangle_{z+dz} = \langle \rho \rangle_z - \langle \rho \rangle_{z+dz} - \frac{2f\rho \langle v_z \rangle^2 dz}{dh}$$

Take lim (dz → 0)

$$\frac{d}{dz} \langle \rho v_z^2 \rangle = -\frac{d \langle p \rangle}{dz} - \frac{2f\rho \langle v_z \rangle^2}{dh} \quad (224)$$

*** Macroscopic Energy Balance

*Ref. 196- Energy Balance per unit volume in absence of convection. In presences of convection

$$\underline{q} = -Ke \nabla T + \sum_{j=1}^s J_{j,e} \bar{H}_j + \bar{C}_p T \underline{V} \quad (225)$$

Substitute in (196)

$$\bar{C}_p \frac{\partial T}{\partial t} + \underline{V} \cdot \underline{q_{conv}} = -\underline{V} \cdot \underline{q_{cond}} + \sum_{i=1}^m (-\Delta H_i) r_i$$

Integrate

$$\frac{d}{dt} \int_{\Omega} \bar{C}_p T dV + \oint_{\partial\Omega} \bar{C}_p T \underline{V} \cdot dA = - \int_{\partial\Omega} \underline{q_{cond}} \cdot dA + \int_{\Omega} \sum_{i=1}^m (-\Delta H_i) r_i dV$$

For perfectly mixed reactor, T is uniform in Ω , such that

$$\frac{d}{dt} (\bar{C}_p T V) + T_f \oint_{\partial\Omega_{in}} \bar{C}_p \underline{V} \cdot dA + T \oint_{\partial\Omega_{out}} \bar{C}_p (\underline{V} \cdot dA) = -U A_n (T - T_c) + V \sum_{i=1}^m (-\Delta H_i) r_i$$

If hold up volume V is constant

$$(\bar{C}_p V) \frac{dT}{dt} - q \bar{C}_p (T_f - T) = -U A_n (T - T_c) + V \sum_{i=1}^m (-\Delta H_i) r_i (C, T) \quad (226)$$

For tubular reactor, using an argument very similar to that used for mass species, we may write axial dispersion model

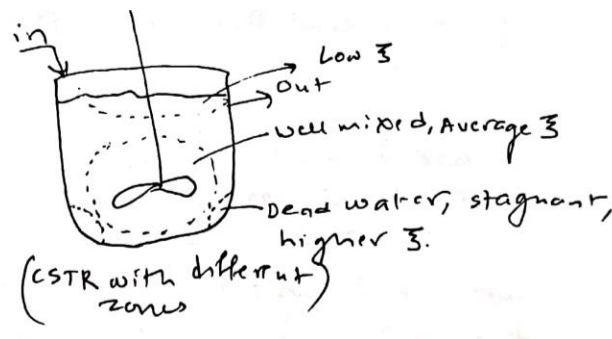
$$\bar{C}_p \frac{\partial T}{\partial t} + \bar{C}_p \langle v_z \rangle \frac{\partial \langle T \rangle}{\partial z} = k_T \frac{\partial^2 \langle T \rangle}{\partial z^2} + \sum_{i=1}^m (-\Delta H_i) r_i - \frac{4U}{dh} (\langle T \rangle - T_c) \quad (227)$$

$$k_T = D_T \bar{C}_p$$

Dankwert type B.C.s also apply here.

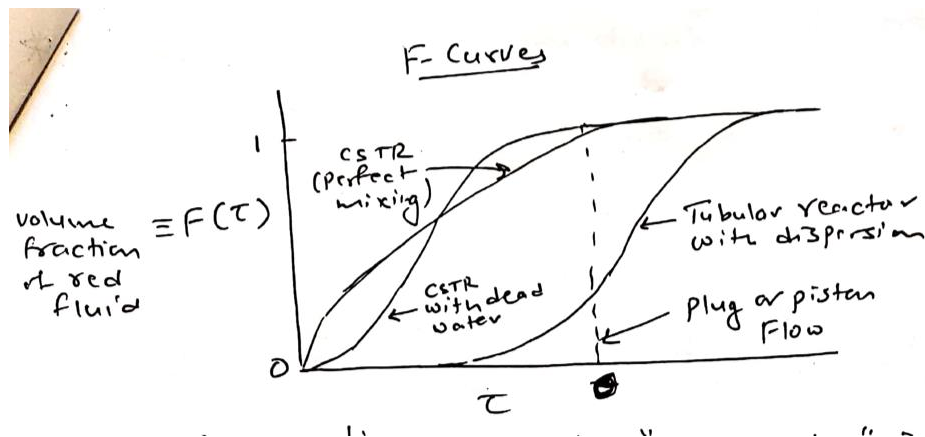
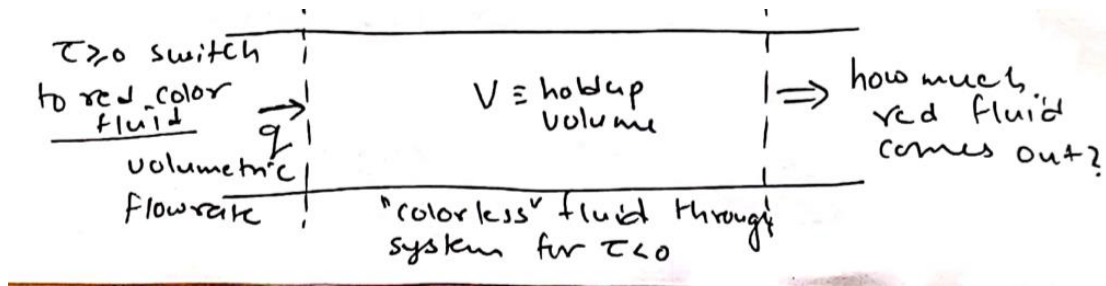
***Residence time distribution

- *In practice, no reactor is perfectly mixed
- *CSTR and well-stirred batch reactors are ideal reactors
- *There may be parts of fluid “held over” in the reactor for a considerable period of time (referred to as “dead water”) before it emerges out of reactor.
- *Danckwerts (1953) introduced a concept of “residence time” for various fluid fractions in the reactor as a result of imperfect mixing.



*Danckwerts's F-curves

- *Consider a continuous reactor with a fully developed flow of colorless liquid
- *At $\tau=0$, we switch the feed from colorless fluid to, say, a red fluid.
- *We now measure the amount of red fluid emerging out of reactor at $\tau>0$



*Average time spent by “dead water” is typically higher than average residence time

$$\frac{V}{q} = \theta \text{ (nominal holding time)}$$

*Define

$I(\tau)d\tau$ = volume fraction of red fluid that has been inside the reactor between τ and $\tau + d\tau$

$F(\tau)$ = volume fraction of red fluid in the effluent after τ units of time

*The volume balance of red fluid can be written as

$$q\tau = V \int_0^\tau I(\tau')d\tau' + \int_0^\tau qF(\tau')d\tau' \quad (228)$$

Differentiate w.r.t τ

$$q = V I(\tau) + qF(\tau)$$

$$I = \theta I(\tau) + F(\tau)$$

$$I - F(\tau) = \theta I(\tau) \quad (229)$$

*Define, residence time distribution (RTD) = $E(\tau)$

$E(\tau)d\tau$ = volume fraction of effluent that has spent time between τ and $\tau + d\tau$

$$= F(\tau + d\tau) - F(\tau)$$

$$E(\tau) = \frac{dF}{d\tau} = -\theta \frac{dI}{d\tau} \quad (230)$$

Case 1:

Average residence time for any flow reactor

$$\bar{\tau} = \int_0^{\infty} \tau E(\tau) d\tau = -\theta \int_0^{\infty} \tau I'(\tau) d\tau = -\theta \int_0^{\infty} I(\tau) d\tau = \theta \quad (231)$$

For any reactor, the average residence time is nominal holding time

Case 2: Perfectly mixed reactor

$E(\tau)$ is complementary to $I(\tau)$

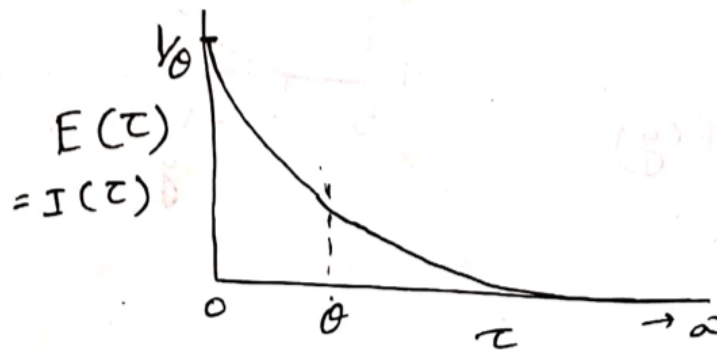
$E(\tau)$ is the residence time distribution for liquid exiting the reactor

$I(\tau)$ is the residence time distribution for liquid in the reactor

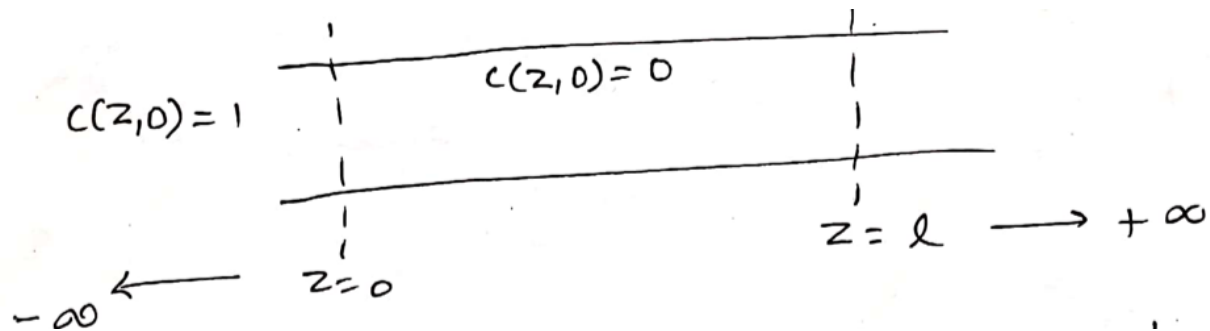
For CSTR, $I(\tau) = E(\tau)$ internal and external residence time distribution are same.

From (230)

$$E(\tau) = -\theta \frac{dI}{d\tau} = -\theta \frac{dE}{d\tau} = \frac{1}{\theta} e^{-\frac{\tau}{\theta}} \quad (232)$$



Case 3: Tubular reactor with Axial Dispersion



*Consider a tubular reactor where tracer is introduced at $z=0$. Assume that tracer has same dispersion coefficient in the reactor as well as outside reactor.

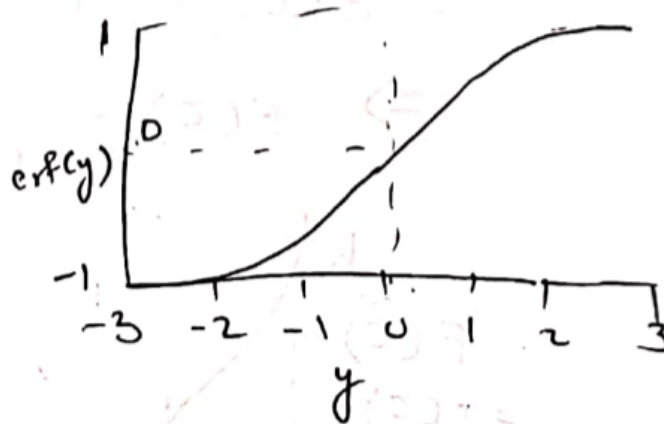
*The dispersion equation for the tracer is given by

$$D \frac{\partial^2 C}{\partial z^2} - v_z \frac{\partial C}{\partial z} = \frac{\partial C}{\partial \tau}, -\infty < z < \infty, \tau > 0 \quad (233)$$

*Although the reactor lies between $z=0$ and $z=u$, we have written (233) for the entire infinite section because the D is the same everywhere.

$$\text{erf}(y) = \frac{1}{\sqrt{\pi}} \int_{-y}^y e^{-t^2} dt$$

$$\text{erf}(y) = 1 - \text{erf}(y)$$



Initial condition

$$C(z,0) = 1, -\infty < z < 0$$

$$C(z,0) = 0, 0 < z < \infty \quad (234)$$

Boundary conditions

$$z \rightarrow -\infty, C \rightarrow 1; z \rightarrow +\infty, C \rightarrow 0 \quad (235)$$

*C is the fraction of tracer, we note that $F(\tau) = C(l, \tau)$

*To solve PDE (233), we introduce transformation

$$x = -xv_z\tau ; u(x, \tau) = C(z, \tau) \quad (236)$$

which yield transformed (233) (234) (235)

$$D \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial \tau} \quad (237)$$

$$u(x,0) = 1, -\infty < x < 0$$

$$u(x,0) = 0, \infty > x > 0 \quad (238)$$

$$x \rightarrow -\infty, u \rightarrow 1 ; x \rightarrow +\infty, u \rightarrow 0 (\tau \geq 0) \quad (239)$$

The general solution of (237) is

$$u(x, \tau) = A + B \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) \quad (240)$$

Apply (238) (239)

$$u(x, \tau) = \frac{1}{2} \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) \quad (241)$$

$$F(\tau) = C(l - v_z\tau, \tau) = \frac{1}{2} \operatorname{erfc}\left(\frac{l - v_z\tau}{2\sqrt{Dt}}\right) \quad (242)$$

*We shall now show how residence time distribution is used to calculate the conversion in reactor.

*Consider $\xi(t)$ as the extent of reaction in a fluid element.

*The average conversion for a reactor with residence time distribution $E(\tau)$ is

$$\bar{\xi} = \int_0^\infty \xi(t) E(\tau) d\tau$$

*If CFD simulations are used then RTD are not required.